SIMATS ENGINEERING

ASSESSMENT TOOL - 3

APPLICATION BASED PROBLEMS

COURSE NAME: Applied Mathematics
COURSE CODE: UBA06

CO 2: Apply Fourier series for a given function by distinguishing even and odd functions.

Bloom's Levels Covered: L4 – Analyze Assessment Tool Weightage: 10%: Total Marks: 50

<table>
<tr><th>Q. No</th><th>Question</th><th>Bloom's Level</th><th>Marks</th></tr>
<tr><td>1</td><td>A rotating turbine blade undergoes periodic displacement during one complete revolution. The displacement measurements are:
<table><tr><td>θ (°)</td><td>0</td><td>45</td><td>90</td><td>135</td><td>180</td><td>225</td><td>270</td><td>315</td></tr><tr><td>y (mm)</td><td>1.0</td><td>1.8</td><td>2.5</td><td>3.0</td><td>2.4</td><td>1.7</td><td>1.2</td><td>0.9</td></tr></table>1. Determine the mean value.
2. Calculate the first and second harmonic coefficients.
3. Formulate the Fourier series up to the second harmonic.
4. Identify the dominant vibration component and suggest corrective measures.</td><td>L4</td><td>10</td></tr>
<tr><td>2</td><td>A thin metallic fin of length $L = \pi$ has the temperature distribution $T(x) = 120x(\pi - x)$, $0 < x < \pi$. Represent the temperature distribution using a half-range sine series.
1. Determine the first four sine coefficients.
2. Approximate the temperature using the first three terms.
3. Explain why the sine series is suitable for this boundary condition.</td><td>L4</td><td>10</td></tr>
<tr><td>3</td><td>The hourly wind speed recorded at a coastal wind farm is given below:
<table><tr><td>Time (h)</td><td>0</td><td>3</td><td>6</td><td>9</td><td>12</td><td>15</td><td>18</td><td>21</td></tr><tr><td>Speed (m/s)</td><td>5</td><td>7</td><td>10</td><td>13</td><td>11</td><td>9</td><td>6</td><td>4</td></tr></table>1. Perform harmonic analysis of the data.
2. Estimate the principal periodic components.
3. Discuss how harmonic analysis assists wind power prediction.</td><td>L4</td><td>10</td></tr>
<tr><td>4</td><td>The initial displacement of a stretched string fixed at both ends is $y(x) = x(\pi - x)^2$, $0 < x < \pi$. Obtain the half-range sine series.
1. Calculate the first three Fourier coefficients.
2. Construct the Fourier series approximation.
3. Explain the relationship between the coefficients and vibration modes.</td><td>L4</td><td>10</td></tr>
<tr><td>5</td><td>The temperature distribution along an insulated plate is $T(x) = 100(1 - x^2/\pi^2)$, $0 < x < \pi$. Represent the function using a half-range cosine series.
1. Determine the first four cosine coefficients.
2. Construct the approximate temperature distribution.
3. Justify why the cosine series is appropriate for the insulated plate.</td><td>L4</td><td>10</td></tr>
</table>

Code of Conduct: I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Excellent (2 Marks)	**Good (1.5 Marks)**	**Satisfactory (1 Mark)**	**Poor (0.5 Mark)**
Evaluating the ability to organize the provided periodic signal values into a format suitable for harmonic analysis.	Minor errors in the organization of tabular data for analysis.	Partial organization of data; difficult to correlate with calculations.	Incorrect or failed extraction of data from the provided set.
Evaluating the correct application of summation formulas for Fourier coefficients $a_0,\ a_n\ and\ b_n$	Mostly correct application of summation formulas with minor errors.	Partial application of formulas; significant gaps in the methodology.	Incorrect or no application of harmonic analysis formulas.
Evaluating the precision of numerical calculations for $a_n\ \ and\ b_n$	Minor numerical errors that do not fundamentally change the result.	Multiple numerical errors affecting the final Fourier coefficients.	Calculations are completely incorrect.
Evaluating the student's ability to conclude which harmonics dominate the signal based on calculated amplitudes and Parseval's identity.	Correct interpretation with minor gaps in the justification or power calculation.	Limited or unclear interpretation of the spectrum and harmonic power.	Incorrect or no interpretation of harmonic dominance.